Generating Webpages with AI for Interactive Education

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Experience will make you improve, but also to find others that can give you good advice. AI?

From Ideas to Interactive Webpages

The 4-Step Workflow

- **Phase 1: Define the Goal**
  What do you want to do?
- **Phase 2: Prepare Content**
  Take 10 minutes to write out your ideas. *How* do you want it to be?
- **Phase 3: Generate & Tweak**
  Paste your prompt into an LLM (like chatGPT, Claude, Gemini, etc). Give feedback to fix bugs or change colors (e.g., "Make the text larger").
- **Phase 4: Publish & Share**
  Show it to other people, keep playing with it.

link: **gemini created this webpage with some advices**

Troubleshooting

- **Expect Glitches:** If something breaks, just tell the AI: *"The [element] isn't working, please fix the bug."*
- **Stay in One Thread:** Don't refresh the LLM page; keep the conversation going so the AI maintains context.
- **Ask for Options:** Not a fan of the look? Ask for 3 different color palette options to choose from.
- **Pro Tip:** Ask the AI to "Audit this code for mobile responsiveness" to ensure it works on tablets and phones.

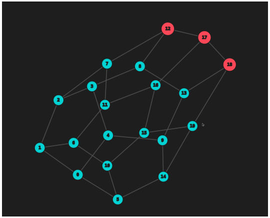
Level Up: The Professional Developer Workflow

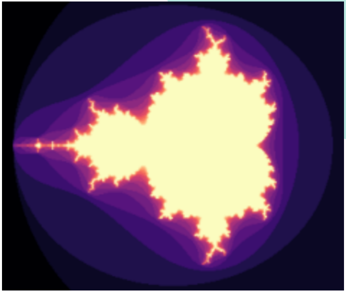
Move beyond the chat box. To build high-quality, permanent educational tools, adopt the industry-standard workflow:

- **The Editor: VS Code (Visual Studio Code)**
  - Install the **Live Preview** extension to see changes instantly.
  - Use **GitHub Copilot** or **Cursor** (an AI-first version of VS Code) to write and debug code directly in your files.
- **Version Control: GitHub**
  - Create a repository (folder) for your project.
  - If a new AI update breaks your quiz, you can "time travel" back to the working version.
- **The Stage: Vercel or GitHub Pages**
  - **Vercel/Netlify:** Connect your GitHub; every time you update your code, your website updates automatically.
  - **GitHub Pages:** The easiest way to host static classroom tools for free under your own name (e.g., `teachername.github.io`).

Two examples from the previous presentation

- Fractals: colab
- Drawing graphs: webpage





(slides of previous presentation)

More Examples

- https://emersonjleon.github.io/emersonjleon/arab/tunisian.html
- https://emersonjleon.github.io/emersonjleon/algoritmos/generador-de-paisajes/2/paisajes3.html
- https://emersonjleon.github.io/emersonjleon/algoritmos/multiplication/multiplications.html
- https://emersonjleon.github.io/emersonjleon/algoritmos/regladelogaritmos/regla.html
- https://emersonjleon.github.io/emersonjleon/games/queensbattle.html
- https://emersonjleon.github.io/emersonjleon/edu/lectura1.html

Final remarks

- Enjoy exploring: How deep the rabbithole goes?
- Keep improving, practice will make you better.
- Long term results vs. short term results. Quality stays longer, but we don't have to make it perfect now. We can do it better later.

(QR link for this presentation)